Note on Commutative Algebra

Jiawen Huang

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Acknowledgement

This note is basically a copy of Eisenbud's Commutative Algebra with a View Toward Algebraic Geometry, with some additional explanations and examples from Eisenbud's lectures and my own understanding.
This latex style is made by Evan Chen.
In this note, all rings R are commutative with unity unless otherwise specified.

January 17, 2026

Corollary 0.1 (Summand of polynomial rings over fields is finitely generated)
Let k be a field, and let R be a k-subalgebra of $S = k[x_1, \ldots, x_n]$. R is a direct summand of S in the sense that there is a surjective R-algebra homomorphism $\phi : S \to R$ such that $\phi|_R = \mathrm{id}_R$ and ϕ preserves degrees. Then R is finitely generated as a k-algebra.

Proof. Intuition: R itself is a graded k-algebra, with grading inherited from S. As a graded algebra, R can be decomposed into its homogeneous components. R_0 is easy to get, just include 1 in the generators. Therefore, to generate R, we really need to generate the positive degree part. That motivates us to look at the ideal R_+.

But R_+ may not be finitely generated as an ideal. However, we know that S is a Noetherian ring, so we can look at the ideal R_+S. It must be finitely generated, say by $f_1, \ldots, f_m \in R_+$.

Now, we claim that $R = k[1, f_1, \ldots, f_m]$. Certainly, the right hand side is contained in the left hand side. For the other direction, it suffices to show that any homogeneous element of R can be generated by $f_1, \ldots, f_m$ with coefficients in k. We prove this by induction on degree d. $d = 0$ is trivial. Now suppose we have shown this for degree less than d. Take any homogeneous element $r \in R$ of degree d. Since $r \in R_+S$, we can write $r = \sum g_i f_i$ for some homogeneous element $g_i \in S$. Apply ϕ to both sides, we get

$$r = \phi(r) = \sum \phi(g_i)\phi(f_i) = \sum \phi(g_i) f_i.$$

Note that $\deg(\phi(g_i)) = \deg(g_i) < d$, so by the inductive hypothesis, each $\phi(g_i)$ can be generated by $f_1, \ldots, f_m$ over k. Therefore, r can also be generated by $f_1, \ldots, f_m$ over k. This finishes the proof. $\square$

Remark 0.2. This shows that the invariant ring of a linearly reductive group acting on a polynomial ring over a field is finitely generated. If the group G is finite, the surjective homomorphism ϕ can be taken as $\phi(f) = \sum_{g \in G} gf$. (Weyl's averaging trick)

Remark 0.3. Graded rings are great because it allows us to use induction on degree. So we only need to worry about homogeneous elements.

January 20, 2026

First day of Commutative Algebra course!

We discuss the history of Commutative Algebra and three of the four big theorems.

Theorem 0.4 (Hilbert's Basis Theorem)
If R is a Noetherian ring, then the polynomial ring $R[x]$ is also Noetherian.

Proof. Let I be an ideal of $R[x]$. Our goal is to show that I is finitely generated. We first pick the smallest degree nonzero polynomial f_1 in I. Next, we take f_2 to be the smallest degree polynomial in I which is not in the ideal generated by f_1, that is $f_2 \in I \setminus (f_1)$. Continuing this process, we get a sequence of polynomials $f_1, f_2, \ldots$ in I such that f_n is the smallest degree polynomial in $I \setminus (f_1, \ldots, f_{n-1})$. Clearly we have $\deg(f_{i+1}) \geq \deg(f_i)$ for all i. Since R is Noetherian, the leading coefficients of f_i generate a finitely generated ideal in R. This means that there exists j such that the leading coefficient of f_1 to f_j generate all the leading coefficients of f_i for $i > j$.

Now, we look at f_{j+1}. Its leading coefficient can be generated by the leading coefficients of f_1 to f_j, that is $a_{j+1} = \sum_{i=1}^{j} r_i a_i$ for some $r_i \in R$. Consider the polynomial

$$g = f_{j+1} - \sum_{i=1}^{j} r_i x^{\deg(f_{j+1}) - \deg(f_i)} f_i$$

Since $\deg(g) < \deg(f_{j+1})$ so $g \in (f_1, \ldots, f_j)$. Thus, $f_{j+1} \in (f_1, \ldots, f_j)$, contradicting the choice of f_{j+1}. Therefore, $I = (f_1, \ldots, f_j)$ is finitely generated. $\square$

Theorem 0.5 (Hilbert's syzygy theorem)
If $R = k[x_1, \ldots, x_n]$ is a graded polynomial ring over a field k, then every finitely generated graded R-module $M = \bigoplus_{s \in \mathbb{Z}} M_s$ has a finite graded free resolution of length at most n. That is, the sequence

$$0 \to F_n \to F_{n-1} \to \cdots \to F_1 \to F_0 \to M \to 0,$$

where each F_i is a finitely generated graded free R-module, is exact.

Proof is omitted.

Theorem 0.6 (Hilber's function theorem)

Let M be a finitely generated graded module over the polynomial ring $R = k[x_1, \dots, x_n]$. Then there exists a polynomial $P(t) \in \mathbb{Q}[t]$ such that for all sufficiently large d, we have $\dim_k M_d = H_M(d) = P(d)$.

We will present two proofs of this theorem.

Proof. We will use the Hilbert's syzygy theorem to prove this. By the theorem, we have a finite graded free resolution of M:

$$0 \to F_n \to F_{n-1} \to \cdots \to F_1 \to F_0 \to M \to 0.$$

Thus, we have $H_M(d) = \sum_{i=0}^{n} (-1)^i H_{F_i}(d)$. So we only need to study the dimension of free modules F_i, which is a direct sum of $k[x_1, \dots, x_n]$. Note that $H_R(d) = \dim_k R_d = \binom{n+d-1}{n-1}$ (star and bar counting trick), which is a polynomial in d when d is sufficiently large. Therefore, $H_M(d)$ is also a polynomial in d for sufficiently large d. $\square$

Remark 0.7 (The Euler characteristic)**.** Why does $H_M(d) = \sum_{i=0}^{n} (-1)^i H_{F_i}(d)$ hold? This result is a property of the Euler characteristic. For any complex of finite-dimensional vector spaces

$$V_* : 0 \to V_n \to V_{n-1} \to \cdots \to V_1 \to V_0 \to 0,$$

We define its Euler characteristic to be

$$\chi(V_*) = \sum_{i=0}^{n} (-1)^i \dim_k V_i.$$

There is a nontrivial fact that the Euler characteristic is 0 if the complex is exact. The easiest proof is via the decomposition of exact sequence into short exact sequences and sum up the dimension relations for each short exact sequence.

$$\begin{tikzcd}
 & & 0 \arrow[dr] & & 0 & & \\
 & & & \operatorname{im} d_{i+1} = \ker d_i \arrow[ur] \arrow[dr] & & & \\
M_{i+2} \arrow[rr, "d_{i+2}"] \arrow[dr] & & M_{i+1} \arrow[ur] \arrow[rr, "d_{i+1}"'] & & M_i \arrow[rr, "d_i"] \arrow[dr] & & M_{i-1} \\
 & \operatorname{im} d_{i+2} = \ker d_{i+1} \arrow[ur] \arrow[dr] & & & & \operatorname{im} d_i = \ker d_{i-1} \arrow[ur] \arrow[dr] & \\
0 \arrow[ur] & & 0 & & 0 \arrow[ur] & & 0
\end{tikzcd}$$

Second one is via induction on n.

Proof. $n = 0$ is trivial. Suppose it's true for number less than n. Consider the short exact sequence

$$0 \to (0 : x_n) \to M(-1) \xrightarrow{x_n} M \to M/x_nM \to 0.$$

Notice that M/x_nM is a finitely generated graded module over $k[x_1, \dots, x_{n-1}]$ (x_n acts as zero) and $(0 : x_n)$ is also a finitely generated graded module over $k[x_1, \dots, x_{n-1}]$ (the generators of $(0 : x_n)$ over $k[x_1, \dots, x_{n-1}, x_n]$ also generate $(0 : x_n)$ over $k[x_1, \dots, x_{n-1}]$ since x_n would act as zero). By the inductive hypothesis, both M/x_nM and $(0 : x_n)$ have Hilbert dimensions agree with polynomials when d is large enough. Thus, by the exactness of the sequence,

$$H_M(d) - H_M(d-1) = H_{M/x_nM}(d) - H_{(0:x_n)}(d-1) = \lambda(d).$$

for some polynomial $\lambda(d)$ when d is large enough. Thus,

$$H_M(d) - H_M(0) = \sum_{i=1}^{d} (H_M(i) - H_M(i-1)) = \sum_{i=1}^{d} \lambda(i).$$

Notice that $\sum_{i=1}^{d} i^k$ is a polynomial in d of degree $k+1$. Therefore, $\sum_{i=1}^{d} \lambda(i)$ is also a polynomial in d when d is large enough. Therefore, $H_M(d)$ also agrees with a polynomial when d is large enough. □

Remark 0.8. Is $H_M(d)$ always finite? The answer is yes if M is a finitely generated module over $k[x_1, \dots, x_n]$. This is nontrivial because $H_M(d)$ is considering the dimension of M as module over k, not over R.

Proof. Consider $M' = \bigoplus_{s=d}^{\infty} M_s$. M' is a submodule of M, so it's finitely generated as R-module. Let $m_1, \dots, m_t$ be the generators of M'. Each m_i has degree at least d. So for any element in M_d, it's a linear combination of m_i with coefficients in R of degree at most 0 (otherwise it would have degree greater than d). Therefore, M_d is spanned by $m_1, \dots, m_t$ over k, which is finite-dimensional. □

I came up with another more intuitive proof.

Proof. M is finitely generated as R-module by $m_1, \dots, m_t$. For some degree d, any element in M_d must be a linear combination of m_i with coefficients as the monomials in R of degree at most $d - \deg(m_i)$. There are only finitely many such monomials, so M_d is finite-dimensional over k. □

January 22, 2026

Today we discuss the Hilbert's Nullstellensatz and the equivalence between the category of affine algebraic varieties and the category of finitely generated reduced k-algebras. For further details, please refer to here.

January 27, 2026

There is a readiness exam today.

For problem 3, we have a stronger result. For any R and ideals I and J, we have an isomorphism

$$\frac{R}{I} \otimes_R \frac{R}{J} \cong \frac{R}{I+J}.$$

Proof. We will show a more general result: For any ideal I, we have the exact sequence

$$0 \to I \to R \to R/I \to 0.$$

Tensoring with N, we get the right exact sequence

$$I \otimes_R N \to R \otimes_R N \to (R/I) \otimes_R N \to 0.$$

Notice that the image of $I \otimes_R N$ in $R \otimes_R N \cong N$ is exactly IN. Thus, we have the isomorphism $\frac{N}{IN} \cong (R/I) \otimes_R N$. Taking $N = R/J$, we get

$$\frac{R/J}{I(R/J)} \cong (R/I) \otimes_R (R/J).$$

Note that $I(R/J) = IR/J = (I+J)/J$ (an elementary result, informally $I(R/J)$ is just I/J, but I may not contain J so we just use $I + J/J$ to avoid confusion), so we have

$$\frac{R/J}{I(R/J)} \cong \frac{R}{I+J}.$$

Combining the two isomorphisms, we get the desired result. □

Next, we discuss localization. We essentially invert elements of a multiplicative subset $U \subseteq R$. The key propositions are the following:

1. There is a natural map $\phi : R \to M[U^{-1}]$ given by $r \mapsto \frac{r}{1}$.
2. For any ideal $J \subseteq R[U^{-1}]$, $I^c = \phi^{-1}(I)$ is an ideal of R disjoint from U. (This is true in general for any ring homomorphism ϕ.)
3. For any ideal $I \subseteq R$ disjoint from U, the extension $I^e = U^{-1}I$ is a proper ideal of $R[U^{-1}]$. (if they are not disjoint, then 1 is in the $U^{-1}I$, so it's not proper).
4. For any $J \subseteq R[U^{-1}]$, we have $J = (J^c)^e$.
5. However, the $(I^e)^c = I$ is false, even if $I \cap S = \emptyset$. Example: let $U = \{1, x, x^2, \ldots\}$ and $R = \mathbb{C}[x, y]$, $(xy)^e = (y)$ in $R[U^{-1}]$ and $((xy)^e)^c = (y)$ in R. The correct statement is $(I^e)^c = \{r \in R : \exists u \in U, ur \in I\}$. If I is prime, then $(I^e)^c = I$.
6. This gives us the following bijection of prime ideals between R and $R[U^{-1}]$:

$$\begin{tikzcd} \{\mathfrak{q} \subseteq R[U^{-1}]\} \arrow[r, bend left, "\phi^{-1}(-)"] & \{\mathfrak{p} \subseteq R : \mathfrak{p} \cap U = \emptyset\} \arrow[l, bend left, "U^{-1}(-)"] \end{tikzcd}$$

Other properties of localization that I think are important:

1. The universal property of localization: for any R-module M and N, multiplicative closed set U. If U acts invertibly on N, then any R-module homomorphism $\psi : M \to N$ factors uniquely through $U^{-1}M$. That is, there exists a unique R-module homomorphism $\tilde{\psi} : U^{-1}M \to N$ such that the following diagram commutes:

$$\begin{tikzcd} M \arrow[rr, "\psi"] \arrow[dr, "\phi"'] & & N \\ & U^{-1}M \arrow[ur, "\tilde{\psi}"'] & \end{tikzcd}$$

2. If R is Noetherian, then so is $R[U^{-1}]$. If R is UFD, then so is $R[U^{-1}]$.

3. Localization commutes with direct sums and direct limits because it's a left adjoint functor. It's left adjoint to the forgetful functor from $R[U^{-1}]$-modules to R-modules, forgetting the inverted elements' action.

4. Left adjointness implies that localization is right exact. In fact, localization is an exact functor.

5. Exactness implies that localization preserves finite limits and colimits. In particular, it preserves kernels, cokernels (quotient of modules), finite intersections (fiber product=a finite limit). Left adjointness implies that it preserves colimits (need not to assume finiteness). In general, left exactness preserves finite limits, right exactness preserves finite colimits.

6. Localization is nice with respect to tensor products: Let M and A be R-modules, N be an $R[U^{-1}]$-module, then we have the isomorphism

$$U^{-1}R \otimes_R M \cong U^{-1}M,$$

$$U^{-1}(M \otimes_R N) \cong U^{-1}M \otimes_{U^{-1}R} N,$$

$$U^{-1}(A \otimes_R M) = U^{-1}A \otimes_{U^{-1}R} U^{-1}M.$$

The first isomorphism tells us $(-) \otimes_R U^{-1}R$ is exact, that is $U^{-1}R$ is a flat R-module.

7. Tensor products are associative: for any R-module M, R-S-bimodule N and S-module P, we have the isomorphism

$$(M \otimes_R N) \otimes_S P \cong M \otimes_R (N \otimes_S P).$$

We also discuss some property of tensor products and Hom functors.

1. $(-) \otimes_R M$ is left adjoint to $\mathrm{Hom}_R(M, -)$. Thus, $\mathrm{Hom}_R(M, -)$ is right exact.

2. $h_N(-) = \mathrm{Hom}_R(-, N) : \text{R-Mod}^{\mathrm{op}} \to \text{R-Mod}$ is right adjoint to $h^N(-)^{\mathrm{op}}$: $\text{R-Mod} \to \text{R-Mod}^{\mathrm{op}}$.

January 29, 2026

Some multiplicative closed sets that are often used:

1. $U = R_x = 1, x, x^2, \ldots$ for some $x \in R$. The localization R_x is often denoted as $R[x^{-1}]$.

2. $U = R \setminus \mathfrak{p}$ for some prime ideal $\mathfrak{p}$. $U^{-1}R$ is often denoted as $R_{\mathfrak{p}}$ and called the localization of R at $\mathfrak{p}$.

Example 0.9 (The Ring of Laurent Polynomials)
$k[x]_x = k[x, x^{-1}]$.

Example 0.10
The regular function on an affine open subset $D(f)$ of an affine variety X is given by the localization of the coordinate ring $A(X)$ at f, that is $A(X)_f$.

Example 0.11 (stalk of a variety at a point)
Let $X \subseteq \mathbb{A}^n$ be an affine variety over an algebraically closed field k. For any point $p \in X$, the stalk of X at p (all regular functions that are well-defined locally at p) is isomorphic to the localization of the coordinate ring $A(X)$ at the maximal ideal $\mathfrak{m}_p$ corresponding to the point p. That is,

$$\mathcal{O}_{X,p} = A(X)_{\mathfrak{m}_p} = \{\frac{f}{g} \mid f, g \in A(X), g(p) \neq 0\}.$$

Example 0.12
$(\mathbb{Z}/10\mathbb{Z})_2 \cong \mathbb{Z}/5\mathbb{Z}$.

Proof. $(\mathbb{Z}/(10))_2 = (\mathbb{Z}_2/(10)_2) = (\mathbb{Z}_2/(5)_2) = (\mathbb{Z}/(5))_2 = \mathbb{Z}/5\mathbb{Z}$.
Most equalities are guaranteed by the fact that localization preserves quotients. The last step is because $\mathbb{Z}/(5)$ is a field, so any nonzero element is invertible. □

Theorem 0.13 (Local Global Principle)
$M = 0$ if and only if $M_{\mathfrak{p}} = 0$ for all prime ideals $\mathfrak{p}$ if and only if $M_{\mathfrak{m}} = 0$ for all maximal ideals $\mathfrak{m}$.

Remark 0.14. One way to understand this theorem is through dimension argument in algebraic geometry. The codimension of a point (the largest dimension of objects containing the point) is the same as the height of the corresponding prime ideal, which is the Krull dimension of the localized ring at that prime ideal. If the codimenson of any point is 0, then the variety is empty! So the coordinate ring of the variety is 0.

Another way to understand this theorem is through the stalks of a sheaf. If a ring of regular functions is 0 at every stalk (the small neighborhood of every point), then the only function in the ring is the zero function. So the ring itself is 0.

Theorem 0.15
A sequence of R-modules is exact if and only if the localized sequence at any prime ideal is exact if and only if the localized sequence at any maximal ideal is exact.